

0.1 Section 1.5 - Nested Quantifiers

Let $\mathbb{P}$ be the set of prime integers

$$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$$

With this in mind, we can write the following in two ways:

Consider the statement "for all positive integers n , $6n - 1$ or $6n + 1$ is prime"

1. $\forall n \in \mathbb{Z}^+, 6n - 1 \in \mathbb{P} \vee 6n + 1 \in \mathbb{P}$
2. $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{P}, 6n - 1 = p \vee 6n + 1 = p$

Example 1 (Example of Nested Quantifiers) 1. For all real x there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R}, xy = 1$$

FALSE, since if $x = 0$, then $xy = 0 \neq 1$

2. For all positive real x , there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}, xy = 1$$

TRUE since if $x \in \mathbb{R}^+$, then $x > 0$, and if $y = \frac{1}{x}$, then $xy = x \frac{1}{x} = 1$

Note

Given x , we can find y

y depends on x

But what happens if we switch $\forall$ and $\exists$?

3. $\exists y \in \mathbb{R} \forall x \in \mathbb{R}^+, xy = 1$

FALSE since if $x = 1$, then $y = 1$, but if $x = 2$, then $y = \frac{1}{2}$, and $1 \neq \frac{1}{2}$

Note

There is a single y that works **FOR ALL** x

y does not depend on x

4. For all positive integers a and b , we have $ab > 0$

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, ab > 0$$

Note

This is the same as $\forall a, b \in \mathbb{Z}^+$

TRUE since $a, b \in \mathbb{Z}^+$, then $a > 0$ and $b > 0$, hence $ab > 0$

5. There exists integers a , and b such that $a^2 + b^2 = 464$

$$\exists a \in \mathbb{Z} \exists b \in \mathbb{Z}, a^2 + b^2 = 464$$

$$\exists b \in \mathbb{Z} \exists a \in \mathbb{Z}, a^2 + b^2 = 464$$

TRUE $a = 8, b = 20$,

$$\begin{aligned} a^2 + b^2 &= 8^2 + 20^2 \\ &= 64 + 400 \\ &= 464 \end{aligned}$$

Note

The order of the **SAME** quantifier doesn't matter, only for **DIFFERENT** quantifiers does it matter

Example 2

Rewrite the following using Quantifiers

1. "There exists positive integers a and b such that $a^2 + b^2 = 15625$

(a) Write as Quantifier

$$\exists a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+, a^2 + b^2 = 15625$$

(b) Write the negation

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, a^2 + b^2 \neq 15625$$

Work

$$a^2 + b^2 = 5^6$$

$$a^2 + b^2 = 5^4 \cdot 5^2$$

$$3^2 + 4^2 = 5^2$$

$$5^4(3^2 + 4^2) = 5^4 \cdot 5^2$$

$$5^4 \cdot 3^2 + 5^4 \cdot 4^2 = 5^6$$

$$(5^2 \cdot 3)^2 + (5^2 \cdot 4)^2 = 5^6$$

$$(75)^2 + (100)^2 = 5^6$$

$$\therefore a = 75, b = 100$$

TRUE

There's more examples but I'm just too lazy to write them if you get the chance, they're on page 3 of your notes